Aptech Computer Education Examination

Developing Mobile Applications using Android - Modular Exam Sample 3

Exam Code: MOD-ANDROID-403 | Total Questions: 20 | Total Marks: 20.00 | Passing Marks: 8.00

Question 1: Which Android storage option stores private primitive data in the form of key-value pairs?

a) **Shared Preferences**
b) Internal Storage
c) External Storage
d) SQLite Database

Reference / Note: SharedPreferences allows activities and apps to save and retrieve persistent key-value pairs of primitive data types.

Question 2: Which of the following tag is used for registering a broadcast receiver in AndroidManifest.xml?

a) <service>
b) <provider>
c) **<receiver>**
d) <uses-permission>

Reference / Note: Aptech Semester 2: Android Application Development, Session 2 & Android Manifest Specification: The <receiver> tag registers a broadcast receiver component.

Question 3: Which transition effect in Android Material Design moves Views in or out from the edges of the scene towards the center or vice versa?

a) **Explode**
b) Slide
c) Fade
d) Bounce

Reference / Note: Aptech Semester 2: Android Application Development, Session 16, Page 963: 'Explode moves views in or out from the center of the scene.'

Question 4: Which of the following Android widgets is used to animate between two or more views that are placed on top of each other?

a) **ViewFlipper**
b) ViewSwitcher
c) ViewAnimator
d) ViewPager

Reference / Note: Aptech Semester 2: Android Application Development, Session 5, Page 318: 'ViewFlipper is used to animate between two or more views.'

Question 5: Which of the following statements accurately defines an Android Menu?

a) **It is a UI component that holds several action items to provide additional functionality to an application**
b) It is a background service that processes touch events
c) It is a database table schema for persisting user preferences
d) It is a manifest tag used exclusively for declaring app permissions

Reference / Note: Menus are user interface components that present options and actions without cluttering the main screen layout.

Question 6: **Which of the following is the initial mandatory element or component of an XML document?**

a) **Prolog (<?xml version="1.0" encoding="utf-8"?>)**

b) Root element

c) Schema header

d) Doctype declaration

Reference / Note: The XML prolog specifies the XML version and character encoding before the root element begins.

Question 7: **Which support library must be added to build.gradle in order to use the RecyclerView class in older Android projects?**

a) **com.android.support:recyclerview-v7**

b) com.android.support:support-v4

c) com.android.support:appcompat-v5

d) com.android.support:design-v8

Reference / Note: The RecyclerView widget was originally distributed as part of the v7 Support Library (now AndroidX RecyclerView).

Question 8: **Which setting preference constant indicates that the Android Data Saver mode has been disabled by the user?**

a) RESTRICT_DATASAVER_STATUS_DISABLED

b) **RESTRICT_BACKGROUND_STATUS_DISABLED**

c) RESTRICT_DATASAVER_DISABLED

d) DISABLED_DATASAVER_BACKGROUND

Reference / Note: ConnectivityManager.getRestrictBackgroundStatus() returns RESTRICT_BACKGROUND_STATUS_DISABLED when data saver is off.

Question 9: **Which helper class is designed to offload short-running background operations from the UI thread without managing threads manually?**

a) **AsyncTask**

b) AsyncOffload

c) SyncTask

d) BackgroundRunner

Reference / Note: AsyncTask enables proper and easy use of the UI thread by performing background computations and publishing results to UI.

Question 10: **Which of the following permissions is classified as a Dangerous Runtime Permission in Android 6.0 (API level 23) and above?**

a) INTERNET

b) ACCESS_NETWORK_STATE

c) **WRITE_EXTERNAL_STORAGE**

d) WAKE_LOCK

Reference / Note: Aptech Semester 2: Android Application Development, Session 16, Page 953 & Android Permission Model: WRITE_EXTERNAL_STORAGE belongs to the STORAGE runtime permission group.

Question 11: **Which onboarding pattern model allows the user to customize the user interface before entering the core app experience?**

a) **Self-select model**

b) Quickstart model

c) Top User Benefits model

d) Gesture Education model

Reference / Note: The Self-select onboarding model allows users to personalize topics, UI preferences, or interests upon initial setup.

Question 12: Which transition effect in Android Material Design can be applied to both enter and exit transitions to animate views outward or inward?

a) **Explode**

b) Fade

c) Swipe

d) Slide

Reference / Note: The Explode transition moves views into or out of the scene toward or away from the center of the screen.

Question 13: Which of the following are the most commonly used Preference classes in Android PreferenceScreen layouts?

a) **CheckBoxPreference, ListPreference, and EditTextPreference**

b) ListViewPreference, SharedViewPreference, and SpinnerPreference

c) TextFieldPreference, RadioPreference, and TablePreference

d) SeekPreference, ScrollPreference, and ButtonPreference

Reference / Note: Standard Android preferences include CheckBoxPreference (boolean toggle), ListPreference (choice list), and EditTextPreference (text entry).

Question 14: Which Java collection class allows dynamic resizing and provides indexed access, but its methods are unsynchronized?

a) **ArrayList**

b) Vector

c) HashSet

d) PriorityQueue

Reference / Note: ArrayList provides fast, indexed, resizable array access and is not synchronized, making it standard for Android UI adapters.

Question 15: Which layout attribute is used to define extra margin or whitespace on the right side of a View inside a ViewGroup?

a) **android:layout_marginRight / android:layout_marginEnd**

b) android:paddingRight

c) android:gravity_right

d) android:layout_weight

Reference / Note: android:layout_marginEnd / android:layout_marginRight specifies extra outer space on the right of the view.

Question 16: Which of the following Android UI widgets uses an Adapter to bind and display a scrollable list of items?

a) **ListView**

b) ScrollView

c) TextView

d) FrameLayout

Reference / Note: AdapterView subclasses such as ListView and GridView require an Adapter to bind data from an underlying collection.

Question 17: Which database engine is integrated directly into the Android OS platform for local relational data storage?

a) **SQLite**
b) MySQL
c) PostgreSQL
d) SQL Server Compact

Reference / Note: Android includes built-in support for SQLite databases accessible via SQLiteOpenHelper and SQLiteDatabase.

Question 18: What is the topmost layer of the Android architecture stack that contains user-facing programs like Browser and Contacts?

a) **Applications (System Apps) Layer**
b) Application Framework Layer
c) Android Runtime (ART/DVM)
d) Linux Kernel

Reference / Note: The Applications layer sits at the top of the Android stack and contains core built-in and third-party apps.

Question 19: Which activity lifecycle state spans from when onStart() is called until onStop() is called?

a) **Visible lifetime**
b) Entire lifetime
c) Foreground (Resumed) lifetime
d) Inactive lifetime

Reference / Note: The visible lifetime of an activity happens between onStart() and onStop(), during which the activity is visible to the user.

Question 20: Where are user files stored when saved to external storage on Android devices?

a) **SD card / Emulated shared external storage**
b) Internal app private directory (/data/data)
c) RAM cache
d) SIM card memory

Reference / Note: External storage refers to removable media (such as an SD card) or internal shared/emulated external filesystem.